FEASIBILITY STUDY · WORKING DRAFT · NOVEMBER 2026

Distributed Participatory Economics Protocol

Can councils plan an economy together, without a market and without a central office? A design, the evidence so far, and how we would find out.

Titan Cassini · Post-Capitalist Labs

What is proposed

INSTITUTIONS

Councils plan

Worker and consumer councils propose what they will produce and consume, then revise round by round. Proposals come only from councils.

COORDINATION

Prices inform

Indicative prices, published each round, show social opportunity cost. A facilitation board computes them by public rules and cannot change any plan.

TECHNOLOGY

Software protects

A local-first protocol keeps each council's state on its own devices, makes tampering detectable, and keeps household requests private.

The economic model is Albert and Hahnel's participatory economics. DPEP asks whether its coordination can actually run, and how to test that.

WHY IT MATTERS

“There is no third way.” Either negotiated market contracts, or binding instructions from planning offices.

Alec Nove (1983), as quoted by Albert and Hahnel (1991)

Participatory planning claims a third way: decentralized, democratic, and non-market. DPEP is an attempt to make that claim testable.

Three kinds of evidence

Demonstrated

Code or data audited

- Convergence in smooth Cobb–Douglas economies
- Failure on Scarf’s classic cycle
- Measured sizes for private household reports

Estimated

Analytical, not yet built

- Round latency and archive sizes
- Phone bandwidth per round
- Costs and staffing by phase

Open

Research targets

- Convergence with indivisible plants
- Whether people sustain the process
- Private checks of household budgets

Every claim in the study carries one of these labels (Table 1.1; Appendix A).

Does planning converge?

11.85

rounds on average to bring every imbalance under 5% in smooth Cobb–Douglas economies (40 large runs of pequod-cljs).

10.54%

floor the same price rule never gets below on Scarf's classic economy, where incomes move with prices.

Round 9

reaches 5% when the same consumers get fixed budgets instead: the instability comes from income feedback.

Open: indivisible plants, Leontief inputs, and incomes that respond to prices. That is what Kill Gate 1 tests.

Source: Table 1.1 and §4.4.3; scripts in research/analysis/ (verify_r2.py, verify_economics.py).

Not a blockchain

Planning is a round-based computation among councils, not millions of transactions per second. So DPEP keeps state with the councils and records only what must be shared.

Each round ends in a small checkpoint that fixes its history, run by a fault-tolerant committee or a witnessed log. Both are tested at Kill Gate 2.

LAYER 4

People deliberate; solvers run on their devices

LAYER 3

Council and household state, local-first and offline

LAYER 2

Federations aggregate; anyone can recompute totals

LAYER 1

Per-round checkpoints and a deterministic price engine

What the design can and cannot promise

Guaranteed

by cryptography, under stated assumptions

- Private channels and reports
- Signed, attributable proposals
- Proof a proposal was counted
- Same prices on every node

Detected

not prevented

- Omitted or altered proposals
- Equivocation by validators
- Gross aggregate overspend
- Duplicate reports

Institutional

rests on people and audit

- Physical output and emissions
- Labor rosters
- Key custody
- Consequences of faults

“Trustless” overstates it: the aim is that no single operator can seize or quietly edit the plan (Fig. 7.5).

Household privacy

Each household's request is split into two random-looking shares, sent to two aggregators run by different operators. Only the neighborhood total is ever reconstructed.

Built on Prio3, an IETF aggregation standard, with an existing server (DAP) for pilots.

PRIVACY

Holds if at least one aggregator is honest

CORRECT TOTALS

Need both aggregators to be honest

STILL OPEN

Proving privately that each request stays within budget

Ready for quantum computers, without the cost

A phone checking one round's checkpoint downloads

290 kB
if every signature were post-quantum

5.5 kB
under the tiered policy

- Confidentiality is post-quantum now (hybrid ML-KEM)
- Long-lived records are signed hybrid
- Per-round objects are signed classically
- Each round's history is anchored post-quantum by several parties

Source: §7.2.1 and Table 7.10; sizes from verify_signature_suites.py. Proposed, for author ratification.

Four architectures, scored honestly

Architecture	Baseline score	Role
D · four-layer hybrid (DPEP)	4.07 of 5	Target design
B · pure peer-to-peer	3.59 of 5	Not chosen
A · federated server (SQL/REST)	3.10 of 5	First pilots
C · per-transaction blockchain	2.54 of 5	Rejected

A simple federated server overtakes D once simplicity, throughput, and cost carry 63% or more of the weight, so it is the right tool for the human probe and the earliest pilots.

Source: Table 6.1 and §6.1.1 (weighted 14-criterion comparison; expert judgments, not measurements).

Four phases, each ending in a test

Phase	Staff and time	Estimate	Ends at
1 · Economic simulator	1.0 FTE · 4–6 months	$41k–$93k	Kill Gate 1
1H · Human probe (parallel)	1.0 FTE · 3–4 months	$38k–$85k	Kill Gate 1H
2 · Distributed testbed	2.5 FTE · 6–8 months	$152k–$305k	Kill Gate 2
3 · Cryptography, mobile, co-op pilot	4.5 FTE · 9–12 months	$495k–$1.07M	Kill Gate 3

$0.73M–$1.55M for the whole program, if every gate passes. Spending stops where a gate fails.

Source: Table 1.2. Assumes $120k–$180k per FTE-year, a planning assumption, not a sourced rate.

Gates that can fail

Every gate has pass and fail conditions, registered before the run, and a stated consequence.

Proceed

All conditions pass

Narrow

Continue only for the cases that passed

Redesign

Rerun once on a fresh registered set

Stop

Publish the negative result; fund no later phase

Example, Gate 1: plans must be *cleared*, not merely feasible. A “feasible” plan once left 42% of output unused.

What a working demonstration would show

Scene	What you see	Phase	Status
Councils plan	20 workplaces and 20 consumer councils revise until the gap closes	1, 1H	Shown for smooth economies
A council goes offline	It keeps working; its proposal counts or is carried forward	2	Specified, not built
An operator cheats	An altered or dropped proposal is exposed and challenged	2	Specified, not built
Private household requests	Two operators compute the total; neither sees a request	3	Measured on a desktop

A target, not software that exists. Each scene has a pre-registered pass and fail condition (§9.2, Table 9.3).

What remains open

Convergence

With indivisible plants, Leontief inputs, and incomes that respond to prices.

Private budget checks

Proving each household stays within budget, privately, when prices change between rounds.

Privacy at the point of use

Collecting goods through the year privately, with offline double spending bounded.

Sustained participation

Whether ordinary people keep planning, round after round, without an expert elite forming.

One report per person

Enforcing it without a population register.

A fair lottery

Public randomness for sortition that no one can bias.

THE PROPOSAL

Not another argument for participatory economics, but a test of its coordination mechanisms, with defined ways to find out that they don’t work.

Start with the cheapest decisive tests: the simulator and the human probe.

github.com/Post-Capitalist-Labs/distributed-participatory-economics-protocol